

VISION KAN: TOWARDS AN ATTENTION-FREE BACKBONE FOR VISION WITH KOLMOGOROV-ARNOLD NETWORKS

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ABSTRACT

Attention mechanisms have become a key module in modern vision backbones due to their ability to model long-range dependencies. However, their quadratic complexity in sequence length and the difficulty of interpreting attention weights limit both scalability and clarity. Recent attention-free architectures demonstrate that strong performance can be achieved without pairwise attention, motivating the search for alternatives. In this work, we introduce Vision KAN (ViK), an attention-free backbone inspired by the Kolmogorov-Arnold Networks. At its core lies MultiPatch-RBfKAN, a unified token mixer that combines (a) patch-wise non-linear transform with Radial Basis Function-based KANs, (b) axis-wise separable mixing for efficient local propagation, and (c) low-rank global mapping for long-range interaction. Employing as a drop-in replacement for attention modules, this formulation tackles the prohibitive cost of full KANs on high-resolution features by adopting a patch-wise grouping strategy with lightweight operators to restore cross-patch dependencies. Experiments on ImageNet-1K show that ViK achieves competitive accuracy with linear complexity, demonstrating the potential of KAN-based token mixing as an efficient and theoretically grounded alternative to attention. Code: <https://github.com/SeriYann/ViK>

Index Terms— Kolmogorov-Arnold Networks, Attention-free vision backbone, Token mixing.

1. INTRODUCTION

Vision backbones in recent years have been dominated by attention-based architectures. The Vision Transformer (ViT) [1] demonstrated that modeling images as sequences of patches with self-attention can rival or surpass convolutional networks, establishing attention as a general-purpose

visual modeling tool. Subsequent variants further extended this line of work, such as Data-efficient Image Transformer (DeiT) [2], which improved training efficiency and practicality, while Pyramid Vision Transformer (PVT) [3] introduced hierarchical feature maps with spatial reduction, bridging Transformers with multi-scale representations.

Despite their success, attention modules face two major limitations: first, their computational and memory complexity grows quadratically with the number of tokens, making them expensive to scale to high-resolution images [4]; second, the token-to-token affinity matrix in self-attention is not directly linked to semantic or structural cues, limiting interpretability [5]. These limitations have motivated the development of attention-free backbones that retain the Transformer architecture while replacing attention with alternative token mixers. MLP-Mixer [6] demonstrated that channel and token MLPs can achieve competitive performance, while MetaFormer [7] further established that the backbone scaffold, consisting of normalization, token mixers, and channel-wise MLPs, can be effective even without attention mechanisms. Together, these works indicate that strong performance in vision does not inherently require quadratic attention, motivating the search for principled and efficient function-based alternatives.

In parallel, the Kolmogorov-Arnold representation theorem (K-A theorem) [8] offers a powerful theoretical insight: any multivariate continuous function can be represented as compositions of univariate functions. This motivates a function-based perspective for modeling token interactions, in contrast to pairwise similarity or linear projection. Recent advances in Kolmogorov-Arnold Networks (KANs) [9, 10] and their applications in diverse areas [11, 12, 13] have shown strong approximation ability, local interpretability, and adaptability across domains, suggesting a promising direction for designing new backbone modules.

Building on these insights, we introduce **Vision KAN (ViK)**, an attention-free backbone that replaces pairwise attention with a function-based token mixer inspired by the K-A theorem. The central component is MultiPatch-RBfKAN, which integrates patch-wise nonlinear modeling with RBF bases, axis-wise separable mixing, and low-rank

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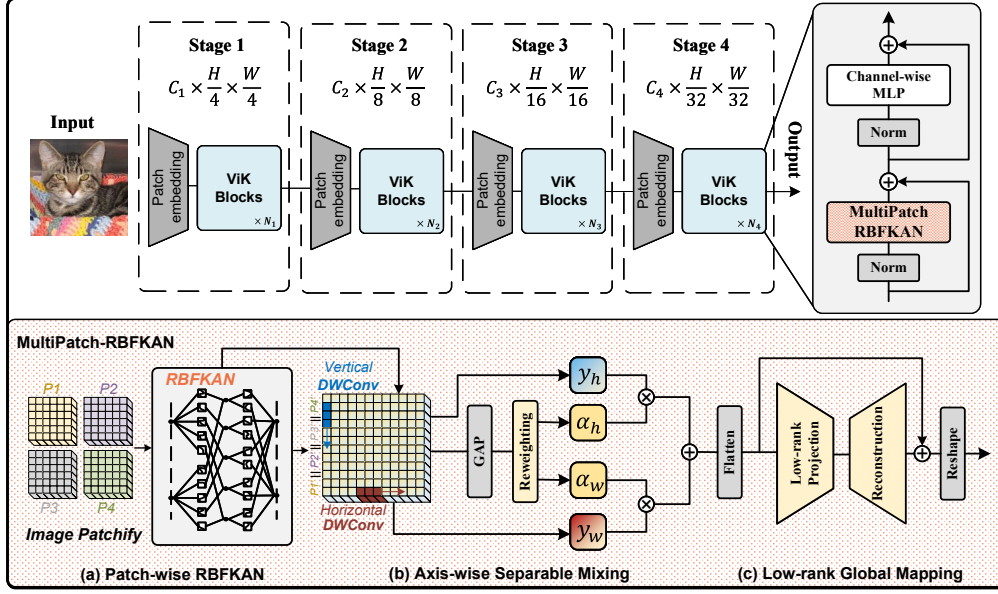


Fig. 1. Overview of the proposed Vision KAN (ViK). The backbone adopts a hierarchical design with four stages, where feature maps are progressively downsampled and processed by ViK blocks. Each ViK block contains the MultiPatch-RBFKAN module, which integrates (a) patch-wise nonlinear modeling with RBFKAN, (b) axis-wise separable depthwise convolutions for direction-sensitive local mixing, and (c) a low-rank global path for efficient long-range dependency modeling. Here, $\oplus$ and $\otimes$ denotes element-wise addition and multiplication, while $\parallel$ denotes concatenation.

global mapping. A key challenge is that directly applying full KAN layers to high-resolution features is computationally prohibitive. We therefore adopt a patch-wise grouping strategy to make KAN feasible at scale. To compensate for the limited cross-patch interaction introduced by grouping, separable mixing enhances local propagation while the global mapping restores long-range dependencies. This unified design balances expressivity and efficiency. Experiments on ImageNet-1K show that ViK achieves competitive accuracy with linear complexity and a principled theoretical foundation.

The key contributions of this work are:

- We propose **Vision KAN (ViK)**, an attention-free vision backbone that replaces self-attention with a KAN-inspired token mixer, offering an efficient alternative to the attention module.
- We design **MultiPatch-RBFKAN**, a unified block that combines patch-wise nonlinear RBF transform, axis-wise separable mixing, and low-rank global mapping to balance expressivity and efficiency.
- Experiments on ImageNet-1K show that ViK achieves competitive accuracy with linear complexity, validating the effectiveness of KAN-based token mixing in large-scale vision tasks.

2. METHOD

2.1. Preliminaries: Kolmogorov-Arnold Networks

The Kolmogorov-Arnold representation theorem states that any multivariate continuous function can be expressed as compositions of univariate functions, i.e., $f(x_1, \dots, x_n) = \sum_q \phi_q \left(\sum_p \phi_{pq}(x_p) \right)$, where $\phi(\cdot)$ are continuous univariate functions. This suggests that complex mappings can be approximated without explicitly modeling pairwise interactions, in contrast to self-attention.

Kolmogorov-Arnold Networks (KANs) implement this principle by replacing fixed activations with parametric basis expansions:

$$\phi(x) = \sum_{j=1}^M w_j \cdot B_j(x), \quad (1)$$

where $B_j(\cdot)$ are basis functions, w_j are trainable weights, and M is the number of basis functions. Compared to conventional MLPs, KANs provide stronger function approximation and local interpretability, since each basis function can be explicitly visualized. In this work, we exploit this property to design an **attention-free vision backbone** that avoids quadratic self-attention yet retains high representational power.

2.2. Overall Architecture

ViK is an attention-free vision backbone with a hierarchical design comprising: a convolutional patch embedding, four stages of ViK blocks with decreasing spatial resolution and increasing channels, and a final normalization and classification head. Each ViK block replaces self-attention with our **MultiPatch-RBFKAN**, a unified token mixer that achieves local-to-global interactions with linear complexity in the number of tokens, serving as a drop-in alternative to quadratic attention.

2.3. KAN-based Token Mixer

Applying a global KAN directly over all tokens is computationally prohibitive. Instead, ViK adopts a structured replacement: **MultiPatch-RBFKAN**, which integrates three complementary operators into a single token-mixing module: (a) a patch-wise KAN for nonlinear local modeling, (b) axis-wise separable mixing to exchange information across patches at linear cost, and (c) a low-rank global mapping to capture long-range dependencies without quadratic overhead.

(a) Patch-wise RBFKAN. Given an input feature map, we first divide it into non-overlapping $p \times p$ patches. Following Eq. 1, where $B(x)$ denotes the basis functions in KAN, we implement $B(x)$ with Radial Basis Functions (RBFs) to model local nonlinear interactions within each patch. Concretely, each patch vector is transformed as:

$$\phi(x) = \sum_{j=1}^M w_j \cdot \exp\left(-\frac{\|x - \mu_j\|^2}{2\sigma_j^2}\right), \quad (2)$$

where μ_j and σ_j are learnable centers and widths, and w_j are trainable scaling weights. Unlike the B-spline basis commonly used in KANs, which requires recursive computation, RBFs allow all basis activations to be calculated in parallel, making them more GPU-friendly and efficient in large-scale vision tasks [14].

(b) Axis-wise Separable Mixing. While patch-wise KAN captures intra-patch dependencies, modeling cross-patch interactions is equally essential. To achieve this efficiently, we apply two depthwise convolutions (DW) along horizontal and vertical axes, followed by global average pooling (GAP) and an MLP-based reweighting:

$$\begin{aligned} \hat{y} &= \alpha_h \cdot DW_h(y) + \alpha_w \cdot DW_w(y), \\ [\alpha_h, \alpha_w] &= \text{Softmax}(f_{\text{MLP}}(\text{GAP}(y))). \end{aligned} \quad (3)$$

This mechanism introduces direction-sensitive local mixing, enabling the model to adaptively emphasize horizontal or vertical dependencies depending on image structures.

(c) Low-rank Global Mapping. Patch-wise local operators are insufficient to capture long-range dependencies. To introduce global context efficiently, we reshape each channel

into a length- N token vector ($N = H \times W$) and apply a low-rank projection:

$$y_{\text{global}} = \mathbf{Q}\mathbf{P}y, \quad \mathbf{P} \in \mathbb{R}^{r \times N}, \quad \mathbf{Q} \in \mathbb{R}^{N \times r}, \quad r \ll N. \quad (4)$$

Here $\mathbf{P}$ compresses spatial tokens into a rank- r latent space, and $\mathbf{Q}$ projects them back, enabling efficient global information exchange.

2.4. Complexity Analysis

Let $N = H \times W$ be the spatial resolution, p the patch size, C the channel dimension, M the number of basis functions, and $r \ll N$ the global rank. The computational cost of each component in MultiPatch-RBFKAN is:

- **Patch-wise RBFKAN:** $\mathcal{O}(N \cdot C \cdot M \cdot F)$, where $F = p^2$ is the patch dimension;
- **Axis-wise Separable Mixing:** $\mathcal{O}(N \cdot C \cdot k)$, with kernel size k ;
- **Low-rank Global Mapping:** $\mathcal{O}(N \cdot C \cdot r)$.

Therefore, the overall complexity per block is

$$\mathcal{O}(N \cdot C \cdot (MF + k + r)),$$

which scales linearly with image size. Compared with quadratic self-attention $\mathcal{O}(N^2 \cdot C)$, this yields a more efficient attention-free alternative while preserving expressive capacity.

3. EXPERIMENTS AND RESULTS

3.1. Experimental Setup

We evaluate ViK on the ImageNet-1K [15] classification benchmark (1.28M training images, 50K validation images, 1K classes). All models are trained from scratch for 300 epochs using AdamW [16] optimizer with weight decay of 0.05 and a peak learning rate of 1×10^{-3} . We adopt the commonly used augmentation and optimization strategy in modern vision backbones, ensuring a fair comparison [3]. We report Top-1 accuracy, number of parameters, and GFLOPs measured at 224×224 input resolution. All experiments are conducted on 4 $\times$ NVIDIA RTX A6000 GPUs.

3.2. Comparison with Representative Vision Backbones

Table 1 compares ViK with a set of widely adopted vision backbone models, including CNN (ResNet [17]), Transformers (ViT [1], DeiT [2], PVT [3]), and MLP-like architecture (ResMLP [18]). Our ViK-Small achieves 76.5% Top-1 accuracy with only 1.6 GFLOPs, providing clear improvements over early Transformers such as ViT-Ti and DeiT-Ti, with only slightly higher computational cost. Compared to

ResMLP-S12 and PVT-Tiny, ViK-Small attains similar or higher accuracy while requiring fewer FLOPs and parameters.

At a larger scale, ViK-Base reaches 80.3% Top-1 accuracy with 3.2 GFLOPs, surpassing ResNet-50 and matching or exceeding the performance of DeiT-Small, PVT-Small, and ResMLP-S24, while maintaining substantially lower complexity. These results demonstrate that ViK achieves a favorable balance between accuracy and efficiency, validating its lightweight design and effective token mixing strategy without relying on attention mechanisms.

Table 1. Comparison between architectures on ImageNet-1K classification.

Model	Type	Params (M)	GFLOPs	Top-1 (%)
ResNet-18 [17]	CNN	11.7	1.8	70.6
ViT-Ti/16 [1]	Attention	5.7	1.3	72.7
DeiT-Tiny [2]	Attention	5.7	1.3	72.2
PVT-Tiny [3]	Attention	13.2	1.9	75.1
ResMLP-S12 [18]	MLP	15	3.0	76.6
ViK-Small (ours)	KAN	13.5	1.6	76.5
ResNet-50 [17]	CNN	25.6	4.1	79.2
ViT-S/16 [1]	Attention	22.1	4.6	78.8
DeiT-Small [2]	Attention	22.1	4.6	79.8
PVT-Small [3]	Attention	24.5	3.8	79.8
ResMLP-S24 [18]	MLP	30	6.0	79.4
ViK-Base (ours)	KAN	24.9	3.2	80.3

Table 2. Ablation study on ViK-Small. “Activation” denotes the type of basis functions in KAN (RBF, B-spline, Wavelet, or replaced by an MLP). “#Basis” indicates the number of basis functions.

Activation	#Basis	Separable Mixing	Global Mapping	Top-1 (%)
RBF	4	✓	✓	74.8
RBF	6	✓	✓	75.7
RBF	10	✓	✓	76.4
B-spline	8	✓	✓	73.8
Wavelet	8	✓	✓	75.2
MLP	-	✓	✓	72.1
RBF	8	✓	✗	74.6
RBF	8	✗	✓	73.9
RBF	8	✓	✓	76.5

3.3. Ablation Study

We conduct ablation experiments on ViK-Small to examine the effect of activation functions, the number of basis functions, and the structural components of MultiPatch-RBfKAN. Table 2 summarizes the results.

Number of basis functions. Increasing the number of RBFs steadily improves accuracy (from 74.8% with 4 bases to 76.4% with 10 bases). Performance peaks at $M=8$ (76.5%) and slightly decreases at $M=10$ (76.4%), indicating saturation with mild overfitting or optimization noise. Given that the computational cost grows linearly with M , we

adopt $M=8$ as the default trade-off between accuracy and efficiency.

Activation functions. Replacing the basis functions in KAN with different alternatives shows that RBF consistently outperforms B-spline and wavelet bases. Moreover, using comparable-size MLP in place of KAN leads to a significant drop (-4.4%). This supports our design choice: RBFs are more GPU-friendly and provide stronger nonlinear approximation in vision tasks.

Separable mixing and global mapping. Removing either axis-wise separable mixing or the low-rank global mapping leads to a clear performance drop (from 76.5% to 74.6% and 73.9%, respectively). Together with the MLP replacement result, these findings validate that our proposed techniques make KAN more suitable for vision tasks by complementing local nonlinear modeling with axis-wise mixing and global interaction.

3.4. Visualization of Learned RBF Mappings

To gain insights into the internal workings of ViK, we visualize the univariate functions $\phi(x)$ learned by RBF across different stages and blocks. Each curve corresponds to a specific input dimension within a patch. The results show that RBF effectively captures nonlinear transformations: functions in shallow stages exhibit more oscillatory shapes sensitive to local variations, while those in deeper stages converge to smoother forms, indicating progressive abstraction toward stable, discriminative features. These univariate mappings provide a clear form of local interpretability.

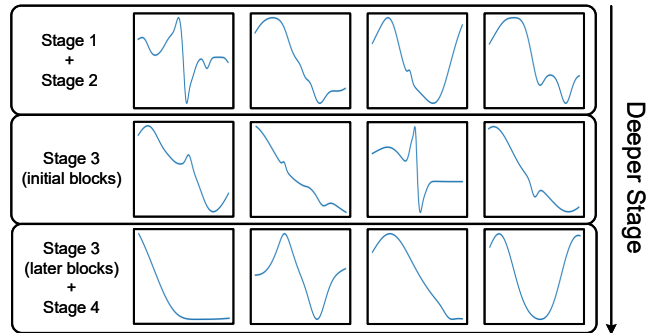


Fig. 2. Examples of the univariate functions $\phi(x)$ learned by RBF across different stages of ViK. Shallow stages exhibit oscillatory nonlinearities, while deeper stages converge to smoother mappings, indicating progressive abstraction.

4. CONCLUSION

We proposed **ViK**, an attention-free vision backbone that replaces quadratic-cost self-attention with a function-based token mixer inspired by the Kolmogorov-Arnold representation theorem. The proposed **MultiPatch-RBfKAN** integrates

patch-wise nonlinear RBF expansions, axis-wise separable mixing, and low-rank global mapping into a unified module, enabling efficient local and global interactions. Experiments on ImageNet-1K demonstrate that ViK achieves competitive accuracy compared with classical attention-based models, while maintaining linear complexity with respect to image size. These results highlight the potential of KAN-based nonlinear transformations as a principled foundation for vision backbones, suggesting a new direction for building efficient architectures in computer vision.

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6. REFERENCES

- [1] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” in *International conference on learning representations*, 2021.
- [2] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou, “Training data-efficient image transformers & distillation through attention,” in *International conference on machine learning*. PMLR, 2021, pp. 10347–10357.
- [3] Wenhai Wang, Enze Xie, Xiang Li, Deng-Ping Fan, Kaitao Song, Ding Liang, Tong Lu, Ping Luo, and Ling Shao, “Pyramid vision transformer: A versatile backbone for dense prediction without convolutions,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 568–578.
- [4] Albert Gu and Tri Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” *arXiv preprint arXiv:2312.00752*, 2023.
- [5] Jasmijn Bastings and Katja Filippova, “The elephant in the interpretability room: Why use attention as explanation when we have saliency methods?,” in *Proceedings of the third blackboxNLP workshop on analyzing and interpreting neural networks for NLP*, 2020, pp. 149–155.
- [6] Ilya O Tolstikhin, Neil Houlsby, Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Thomas Unterthiner, Jessica Yung, Andreas Steiner, Daniel Keysers, Jakob Uszkoreit, et al., “Mlp-mixer: An all-mlp architecture for vision,” *Advances in neural information processing systems*, vol. 34, pp. 24261–24272, 2021.
- [7] Weihao Yu, Mi Luo, Pan Zhou, Chenyang Si, Yichen Zhou, Xinchao Wang, Jiashi Feng, and Shuicheng Yan, “Metaformer is actually what you need for vision,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10819–10829.
- [8] Andreĭ Nikolaevich Kolmogorov, *On the representation of continuous functions of several variables by superpositions of continuous functions of a smaller number of variables*, American Mathematical Society, 1961.
- [9] Ziming Liu, Yixuan Wang, Sachin Vaidya, Fabian Ruehle, James Halverson, Marin Soljacic, Thomas Y Hou, and Max Tegmark, “Kan: Kolmogorov–arnold networks,” in *International conference on learning representations*, 2025.
- [10] Ziming Liu, Pingchuan Ma, Yixuan Wang, Wojciech Matusik, and Max Tegmark, “Kan 2.0: Kolmogorov–arnold networks meet science,” *arXiv preprint arXiv:2408.10205*, 2024.
- [11] Zhuoqin Yang, Jiansong Zhang, Xiaoling Luo, Zheng Lu, and Linlin Shen, “Activation space selectable kolmogorov-arnold networks,” *arXiv preprint arXiv:2408.08338*, 2024.
- [12] Shaode Yu, Ze Chen, Zhimu Yang, Jiacheng Gu, Bizu Feng, and Qiurui Sun, “Exploring kolmogorov-arnold networks for realistic image sharpness assessment,” in *ICASSP 2025-2025 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2025, pp. 1–5.
- [13] Zhuoqin Yang, Jiansong Zhang, Xiaoling Luo, Zheng Lu, and Linlin Shen, “Medkan: An advanced kolmogorov-arnold network for medical image classification,” *arXiv preprint arXiv:2502.18416*, 2025.
- [14] Ziyao Li, “Kolmogorov-arnold networks are radial basis function networks,” *arXiv preprint arXiv:2405.06721*, 2024.
- [15] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei, “Imagenet: A large-scale hierarchical image database,” in *2009 IEEE conference on computer vision and pattern recognition*. IEEE, 2009, pp. 248–255.
- [16] Ilya Loshchilov and Frank Hutter, “Decoupled weight decay regularization,” in *International conference on learning representations*, 2019.

- [17] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
- [18] Hugo Touvron, Piotr Bojanowski, Mathilde Caron, Matthieu Cord, Alaaeldin El-Nouby, Edouard Grave, Gautier Izacard, Armand Joulin, Gabriel Synnaeve, Jakob Verbeek, et al., “Resmlp: Feedforward networks for image classification with data-efficient training,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 45, no. 4, pp. 5314–5321, 2022.



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